

★ WEEK 4 · PART 3 of 3

ARIMA in Practice

Building, Automating, and Honestly Evaluating Forecasts

Transforms · auto_arima · rolling back-tests · the finance caveat

Theory meets reality. Part 3 is the practitioner's playbook: prepare the data, let automation search, validate out of sample, and know exactly where ARIMA belongs — and where it does not.

3

PART 3 · IN PRACTICE

From a Model to a Real Forecast

Data preparation, automated model selection, rigorous out-of-sample evaluation, and a clear map of where ARIMA/SARIMA earns its keep in finance.

Everything, Put to Work

Parts 1–2 built the model; Part 3 uses it well

1

We can specify any ARIMA or SARIMA

Trend via d , season via D , dynamics via (p, q) and (P, Q) . The specification problem is solved — now comes the craft of applying it.

2

Real forecasting is mostly discipline

Prepare the data, let tools search the orders, and — above all — validate on data the model never saw. That discipline is what separates a forecast from a guess.

3

And know the boundaries

ARIMA shines on trending, seasonal, memory-bearing series; it is nearly useless on raw returns. Knowing which is the whole point of this course.

Part 3 in one line: prepare, automate the search, validate out of sample, and respect the limits.

The Complete Pipeline

Weeks 2–4 collapse into one routine

Transform, difference to stationarity, select the orders, diagnose the residuals, then forecast and back-test. Every week of this block is one station on that line.

The stations

- Stationarize: difference for trend (d) and season (D).
- Identify: ACF/PACF or an automated search over orders.
- Diagnose: confirm the residuals are white noise.

The discipline that binds it

- Prefer the simplest model that passes diagnostics.
- Never trust in-sample fit — validate out of sample.
- Compare every forecast against a naive benchmark.

Transform Before You Model

Stabilize the variance first

$$y_t^{(\lambda)} = \begin{cases} \frac{y_t^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln y_t & \lambda = 0 \end{cases} \quad (\text{Box-Cox})$$

When and why

- If seasonal swings grow with the level, take logs first.
- A log turns multiplicative seasonality into additive.
- Box-Cox generalizes the log with a tunable parameter λ .

What it buys

- A roughly constant variance — what ARIMA assumes.
- More symmetric, better-behaved residuals.
- Remember to invert the transform when reporting forecasts.

Outliers and Missing Data

Clean inputs, or clean nothing

Outliers

- A single spike can distort every estimated coefficient.
- Flag it, understand its cause, and consider an intervention term.
- Never delete data silently — document every adjustment.

Missing values & gaps

- ARIMA needs an evenly-spaced, complete series.
- Interpolate short gaps; flag long ones as structural breaks.
- Align the calendar carefully — especially across markets.

Manual vs Automated Selection

Two ways to choose the orders

Hand-reading the ACF/PACF builds intuition; automated search scales it. In practice you let the tool propose, then apply human judgment to the shortlist.

Manual (Box-Jenkins)

- Read the ACF/PACF and propose orders yourself.
- Slow, but it teaches you what the data is doing.
- Essential for learning — and for sanity-checking the tool.

Automated (`auto_arima`)

- Searches many $(p,d,q)(P,D,Q)$ combinations quickly.
- Ranks them by AIC or BIC and returns the best.
- Fast and consistent — but blind to context.

How auto_arima Searches

Automation, demystified

auto_arima runs unit-root tests to pick d and D , then steps through candidate orders, fitting each and keeping the one with the lowest information criterion.

What it automates

- Choosing d and D via ADF/KPSS-type tests.
- A stepwise search over (p, q) and (P, Q) .
- Ranking candidates by AIC or BIC.

What you still steer

- The transform, the season length s , and the search range.
- Whether to allow a trend/drift term.
- The final call, after reading the residuals.

What Automation Cannot Do

The tool proposes; you dispose

Its blind spots

- It optimizes a criterion, not economic sense.
- It can miss structural breaks and one-off events.
- A low AIC does not guarantee good forecasts.

Your irreplaceable role

- Judge whether the model is plausible for the data.
- Read the residual diagnostics it will not read for you.
- Validate out of sample — the criterion is only in-sample.

Comparing Candidate Models

The comparison that actually decides

$$\text{choose model} = \arg \min_{\text{candidates}} \text{out-of-sample error (not in-sample fit)}$$

The right criterion

- AIC/BIC narrow the field quickly and cheaply.
- But the final judge is out-of-sample forecast error.
- A model that fits the past can still forecast poorly.

How to compare fairly

- Score all candidates on the SAME held-out period.
- Include the naive benchmark in the comparison.
- Prefer the simpler model when scores are close.

Forecast Evaluation, Done Right

The out-of-sample rule, made concrete

Split time in order: fit on the past, forecast the future, and score only on data the model never saw. Anything else measures memorization, not forecasting.

The cardinal rule

- Never let future information leak into the fit.
- Respect time order — no shuffling, no peeking ahead.
- Score on the held-out period alone.

Why it is non-negotiable

- In-sample error always flatters complex models.
- Only out-of-sample error predicts real performance.
- This is where most forecasting projects quietly fail.

Train/Test and Rolling-Origin Back-Tests

One split is a start; rolling is rigorous

A single split

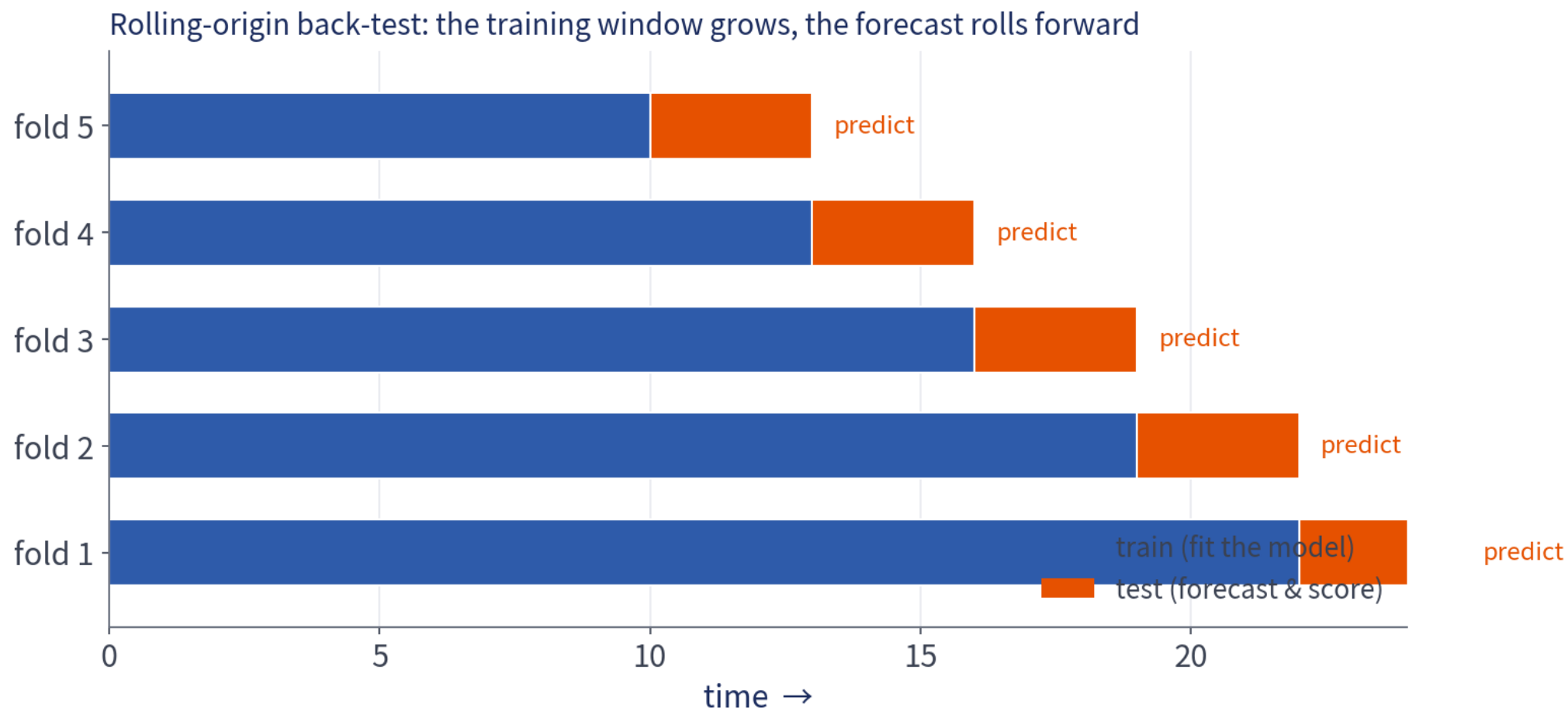
- Fit on the first part, forecast the last part, score it.
- Simple, fast, and far better than in-sample alone.
- But it rests on just one window — possibly lucky or unlucky.

Rolling-origin (walk-forward)

- Repeat the split, sliding the origin forward each time.
- Average the errors across many forecasts.
- A robust estimate of true out-of-sample accuracy.

The Rolling-Origin Back-Test

Fit on the blue window, forecast the orange, score it — then roll the origin forward and repeat. Averaging over folds gives an honest accuracy estimate.



Metrics and the Benchmark

Numbers only mean something in comparison

Element	What it is	Why it matters
RMSE / MAE	average size of the forecast error	the headline accuracy number
Naive benchmark	'no-change' or seasonal-naive forecast	the bar any model must clear
Skill score	improvement over the benchmark	turns raw error into 'is it worth it?'

A small RMSE is meaningless in isolation — always report it beside the naive benchmark's error.

Prediction Intervals and Coverage

Are your 95% bands really 95%?

$$\text{coverage} = \Pr(y_{T+h} \in [\hat{y}_{T+h} \pm z \sigma_h]) \stackrel{?}{\approx} 95\%$$

The honest question

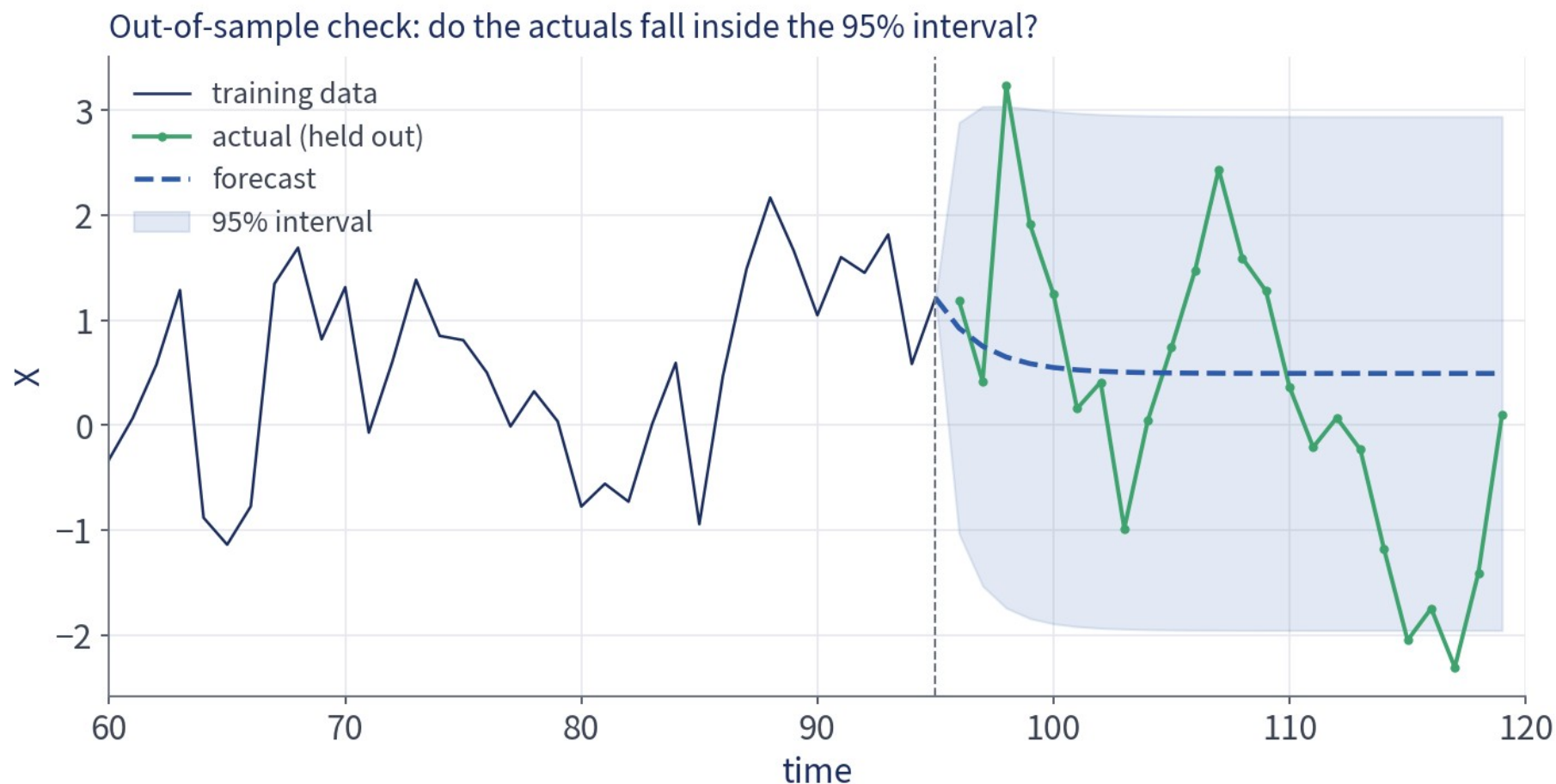
- Over many forecasts, do ~95% of actuals land in the band?
- If far fewer do, your intervals are too narrow.
- Fat-tailed or clustered errors are the usual culprit.

Why it matters most in finance

- Risk decisions hinge on the interval, not the point.
- Under-covered intervals understate tail risk badly.
- Check coverage on the back-test, not just the point error.

Out-of-Sample: Forecast vs Actual

The real test: hold out the recent data, forecast it, and see whether the actuals (green) fall inside the 95% interval. Most should — if the model is honest.



ARIMA on Prices vs Returns

The single most important practical distinction

An ARIMA on a price LEVEL is essentially a random walk with drift — useful for the trend. An ARIMA on RETURNS finds almost nothing, because returns are close to unpredictable.

Price levels ($I(1)$)

- ARIMA models the trend; the forecast continues upward.
- It is effectively a random walk with drift.
- Fine for level projection, weak for timing.

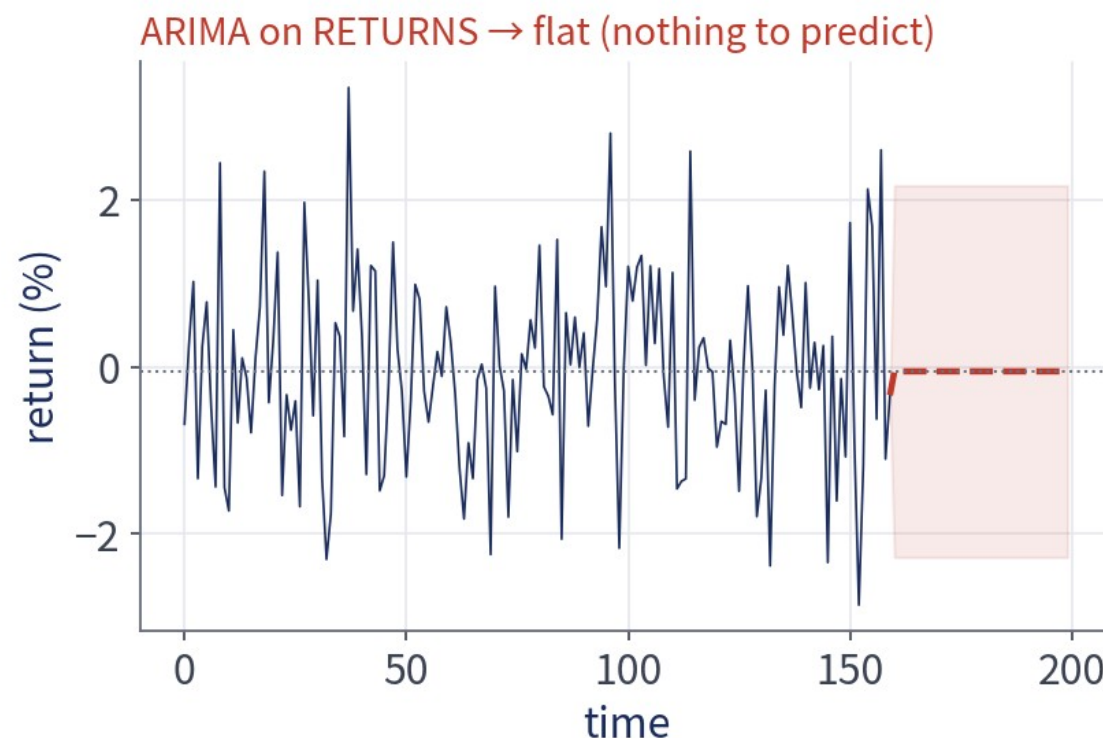
Returns ($\approx I(0)$, $\approx$ white noise)

- The ACF is near zero — no linear structure to fit.
- The forecast collapses to the mean return.
- The action is in the VARIANCE, not the mean.

Why You Model the Level, Not the Returns

Left: an ARIMA on the price level extends the trend. Right: on returns it forecasts a flat mean — there is simply no linear structure to project.

The finance caveat: model the price level, not the (near-random) returns



Where ARIMA/SARIMA Genuinely Applies

A quick fit-for-purpose guide

Domain	Good fit?	Why
Macro series (GDP, CPI)	Yes	trend + memory + clear seasonality
Demand, sales, volumes	Yes	strong, stable seasonal patterns
Asset price levels	Partly	$I(1)$; ARIMA $\approx$ random walk with drift
Asset returns	No	$\approx$ unpredictable; model the volatility instead

Rule of thumb: model levels, macro, and demand; be deeply skeptical of forecasting returns.

When to Move Beyond ARIMA

Three signals the tool has run out of road

The warning signs

- The residual MEAN is clean but its VARIANCE clusters.
- The relationship shifts — a structural break or regime change.
- The dynamics are clearly nonlinear or threshold-driven.

What comes next

- Clustered variance → volatility models (ARCH/GARCH).
- Regime shifts → Markov-switching or time-varying models.
- Nonlinearity → threshold, state-space, or ML approaches.

The Bridge to Volatility Models

Where the returns' structure was hiding

ARIMA models the mean and finds returns unpredictable. But their squared or absolute values are strongly autocorrelated — volatility clusters. That is the doorway to GARCH.

The handoff

- ARIMA/SARIMA: model the level, trend, and season.
- Whatever mean structure exists, it captures.
- What it leaves behind is time-varying variance.

The same tools, one level down

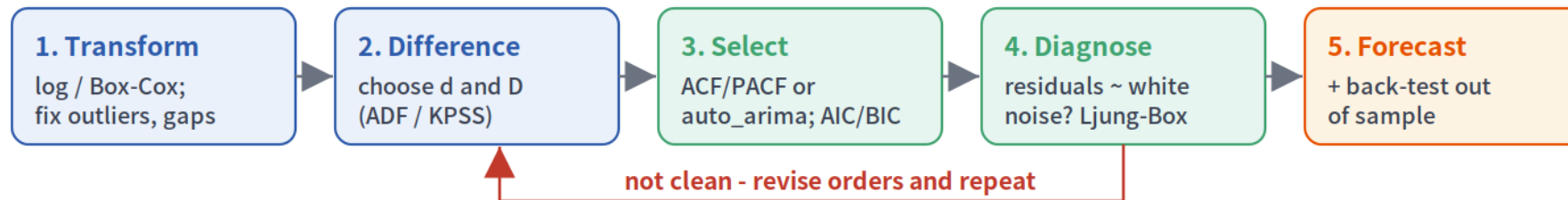
- Apply the ACF logic to squared returns.
- Model the variance as its own autoregressive process.
- That is exactly what GARCH does — the natural sequel.

The Practitioner Pipeline

Transform, difference, select, diagnose, forecast — with a feedback loop when residuals aren't clean, and an off-ramp to volatility models when the mean has nothing left to give.

The ARIMA/SARIMA Practitioner Pipeline

Weeks 2 to 4 as one routine - with an off-ramp to volatility models when the mean has nothing left.



ARIMA / SARIMA is right when...

- the series has a trend and/or a season
- there is genuine linear memory in the mean
- macro, demand, volumes, fundamentals
- residuals become clean white noise

Move beyond it when...

- the mean is ~unpredictable (raw returns)
- residual VARIANCE clusters (volatility)
- relationships shift (regime change)

then the next step is a volatility model (GARCH).

An Automated, Back-Tested Pipeline

Direct Claude Code from raw data to an honest score

GOAL Build an end-to-end pipeline: transform, auto-select the ARIMA/SARIMA, then back-test it against a naive benchmark.

[INPUT] Prompt · Claude Code

Write and run `pipeline_lab.py` that:

1. loads a trending, seasonal series and log-transforms it if the swings grow;
 2. runs `auto_arima` to pick $(p,d,q)(P,D,Q)[s]$;
 3. back-tests with a rolling origin, reporting RMSE for the model AND a seasonal-naive benchmark, plus interval coverage;
 4. plots one out-of-sample forecast vs actual.
- Tell me: does the model beat the benchmark, and are the 95% intervals well-calibrated?

How to work with it

- Sanity-check `auto_arima`'s pick against the ACF.
- Beating a seasonal-naive benchmark is the real win.
- Coverage near 95% means honest intervals.
- Then try it on returns — watch it add nothing.

Deliverable: `pipeline_lab.py` + a back-test report with benchmark and coverage.

Three Questions Before You Finish

If you can answer these, Week 4 has landed

Q1

auto_arima returns the lowest-AIC model. Is that automatically your best forecaster?

No — validate OOS

Q2

Your 95% intervals contain only 70% of actuals in the back-test. What's wrong?

Intervals too narrow

Q3

Should you fit an ARIMA to a stock's daily returns to time trades?

No — returns ~ noise

All three separate a textbook fit from a forecast you could actually trust.

Three Pitfalls to Avoid

Where practical forecasting goes wrong

Common mistakes

- "auto_arima chose it, so it must be right."
- "A great back-test RMSE means a great model."
- "ARIMA on returns will give me a trading signal."

The reality

- Automation optimizes a criterion, not judgment — always sanity-check.
- Compare to a benchmark and check interval coverage, not just RMSE.
- Returns are $\approx$ unpredictable; the edge, if any, is in volatility.

Week 4 in Three Ideas

Pulling the whole week together

1

ARIMA integrates the trend; SARIMA adds the season

Part 1: difference d times, then ARMA. Part 2: do it again at the seasonal lag s . The airline model captures both with just two parameters.

2

Practice is preparation, automation, and honest evaluation

Part 3: transform and clean the data, let `auto_arima` search, then validate with rolling back-tests against a naive benchmark — and check interval coverage.

3

Know the boundary: mean vs variance

ARIMA models the mean and finds little in raw returns. The structure that remains lives in the variance — the doorway to volatility models.

Next — modeling what ARIMA leaves behind: volatility clustering, with ARCH and GARCH.